Mirza Farhan Bin Tarek

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Research Objective

Currently doing a PhD in Computer Science with a focus on the applications of artificial intelligence (AI) and machine learning (ML) in healthcare. Particularly interested in using ML, computer vision and AI to create novel approaches for enhancing patient outcomes, lowering healthcare costs, and creating techniques for quicker, more accurate diagnosis. Dedicated to being an enthusiastic team player who is driven to advance healthcare technologies.

Education

2022 – Present **Ph.D. in Computer Science**, University of Delaware, USA

2017 – 2018 **Individual Modules under Erasmus+ KA1 Mobility Program**, Staffordshire University, United Kingdom
Thesis title: *Spatio-temporal analysis of large air pollution data*.

2014 – 2018 **BSc in Computer Science and Engineering**, United International University, Dhaka, Bangladesh

Employment History

08/2022 – Present **Graduate Teaching Assistant,** Department of Computer and Information Sciences, University of Delaware, US.

06/2023 – 08/2023 **Instructor (On Contract),** Department of Computer and Information Sciences, University of Delaware, US.

05/2019 – 08/2022 **Lecturer,** Department of Computer Science and Engineering, Brac University, Dhaka, Bangladesh.

- Instructed and mentored junior and senior year undergraduate students.
- Led a team of faculty members for the *CSE340: Computer Architecture* and the *CSE360: Computer Interfacing* labs, ensuring smooth and efficient coordination.
- Improved the quality of the curriculum for CSE340 by modifying it to align with outcome-based-education principles.
- Contributed to the development of the *CSE461: Intro to Robotics* curriculum, ensuring that it met the highest academic standards as well as follows the OBE principles.
- Received positive feedback from students and colleagues for my teaching style, organization, and communication skills.

Employment History (continued)

02/2019 – 05/2019 **IT Executive,**, Bangladesh Institute of Journalism and Electronic Media, Dhaka, Bangladesh.

- Managed IT-related tasks such as organizational website management, document management, etc.

Research Publications

Conference Proceedings

1. R. Poulain, M. F. Bin Tarek, and R. Beheshti, "Improving fairness in ai models on electronic health records: The case for federated learning methods," in *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*, ser. FAccT '23, Chicago, IL, USA: Association for Computing Machinery, 2023, pp. 1599–1608, ISBN: 9798400701924. DOI: 10.1145/3593013.3594102.
2. M. F. B. Tarek, M. Asaduzzaman, and M. Patwary, "Spatio-temporal analysis of large air pollution data," in *2018 10th International Conference on Electrical and Computer Engineering (ICECE)*, IEEE, 2018, pp. 221–224.

Skills

Languages	Strong reading, writing and speaking competencies for English, Bengali.
Coding	Python, C, Java, LaTeX
Databases	MySQL, Oracle.
Web Dev	HTML, CSS.
Misc.	Academic research, teaching, LaTeX typesetting and publishing.

Projects

2021 **Skin Cancer Classification with Ensemble of Deep Convolutional Neural Networks**: In this work, deep convolution networks e.g., VGG-16, Efficient NetB3 and ResNet-50 were used to identify skin cancer from dermoscopic images.

2017 **BDDoctors**: An online doctor appointment system with a database. Users can easily book appointments with preferred doctors according to their location, specialty, etc.

Awards, Achievements and Certifications

Awards and Achievements

2018 **Summa Cum Laude**, BSc in CSE, United International University, Dhaka, Bangladesh.

2017 **Participated in the Erasmus+ KA1 mobility program (Funded by Erasmus+ EU)** to study for 6 months at Staffordshire University, Stoke-on-Trent, the United Kingdom in 2017-2018.

Certifications

2020 **Deep Learning Specialization**. Awarded by Coursera.

2017 **Web development using ASP.NET** . Awarded by CDIP (Centre for Development of IT Professionals) ,United International University, Dhaka, Bangladesh.

References

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